

ABDULLAH AL KHALID

Software Engineer / Distributed Systems

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SUMMARY

Software engineering graduate focused on distributed systems, databases, and backend infrastructure. Experience with .NET APIs, SQL data pipelines, and systems work in Rust and Go — including Raft, gossip protocols, CRDTs, and append-only logs.

WORK EXPERIENCE

DKSH

Kuala Lumpur, Malaysia

Software Engineer Intern

Sep 2024 – Feb 2025

- Gained hands-on experience in application management, web development, and automation while collaborating with cross-functional teams in a production environment.
- Contributed to an internal service write-up involving .NET Core for API management and a data pipeline.
- Developed a website development plugin.
- Managed data migration and integration with SQL Server / MSSQL databases.

EDUCATION

Universiti Teknologi Malaysia (UTM)

Skudai, Johor, Malaysia

Bachelor of Computer Science with Honours in Software Engineering

2021 – 2025

ACTIVITIES & COMPETITIONS

RecyclO

Project

Plastic Waste Management System

- Designed a plastic waste management system aimed at reducing plastic waste through practical collection and processing workflows.
- Focused on collaborative, sustainable solutions intended to reduce environmental impact.

Flood Remission System

Project

Disaster Management

- Developed a multipurpose disaster-management application focused on flood mitigation.
- Used real-time weather sensors and satellite imagery to track flood conditions and forecast flooding events.
- Sent early-warning alerts to residents in flood-prone areas and supported authorities in coordinating rescue, relief, and shelter.

PASSION PROJECTS

ironbarrel (Rust)

A log-structured hash table key-value storage engine inspired by the original bitcask

- Built a Rust-based embedded key-value store using append-only log files, in-memory indexing, tombstone-based deletion, TTL expiration, and merge compaction.
- Implemented crash recovery and data-integrity checks with CRC32 validation, hint files, corrupted-record detection, file rollover recovery, and fallback data scanning.
- Added concurrent access and process safety using DashMap, parking_lot, positional file I/O, serialized write transactions, and shared/exclusive OS file locks.
- Developed comprehensive tests and GitHub Actions CI for CRUD operations, concurrency, recovery, corruption handling, compaction, locking, and cross-platform builds.

pkgLog (Go)

Idiomatic log writer for distributed systems

- Implemented an append-only log that stores records in memory and syncs them to persistent storage.
- Used a memory map for indexing and faster caching.

Miserable Messengers / Gossip Gloomers (Rust, Maelstrom/Jepsen)

Fly.io distributed systems challenge series

- Built a distributed systems node engine in Rust to solve the Fly.io Gossip Gloomers challenges.
- Implemented an actor-based channel model for asynchronous I/O.
- Designed custom protocols for anti-entropy, gossip broadcast, CRDTs, and a linearizable key-value store.
- Verified nodes against network splits, packet drops, and concurrent client requests using Jepsen-based Maelstrom tests.

LICENSES AND CERTIFICATIONS

React, Node.js, Express & MongoDB – The MERN Full Stack Guide

Issued by *Maximilian Schwarzmüller & Manuel Lorenz*

AWS Academy Graduate – AWS Academy Cloud Foundations

Issued by *Amazon Web Services (AWS Academy)*

International English Language Testing System (IELTS)

Issued by *IELTS*

COURSES

- **React, Node.js, Express & MongoDB – The MERN Full Stack Guide** – Maximilian Schwarzmüller & Manuel Lorenz
- **Product Management Bootcamp** (HDNB)
- **Learn React** – Bob Ziroll
- **MERN Stack Course – Convert Backend to Serverless with MongoDB Realm** – Beau Carnes

SKILLS, LANGUAGES, INTERESTS

- **Spoken languages:** English (Professional – IELTS)
- **Languages:** Rust, Go, C# / .NET Core, SQL, JavaScript
- **Distributed Systems:** Raft, Gossip / anti-entropy, CRDTs, Linearizable KV stores, Append-only logs, Actor-based concurrency, Maelstrom / Jepsen testing
- **Backend & Data:** REST APIs, SQL Server / MSSQL, Data migration & integration, MongoDB, Node.js, Express
- **Web:** React, MERN stack, Website plugins
- **Infrastructure:** Kubernetes, HashiCorp Consul, AWS Cloud Foundations, Git
- **Interests:** Distributed systems, Databases, Systems programming

REFERENCES

A. S. M. Khairuzzaman

Managing Director & CEO (Acting); Deputy Managing Director & COO

Central Counterparty Bangladesh PLC

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